Machine Markets: Input and Financial-Output Gate Review

Research revision evidence

October 8, 2026

1 A Finite-Window Gate with Explicit Units and Existing Exposure

Specify a scalar executed quantity v_j in declared base units, a fixed-coordinate state, its input map H, and a monetary model-loss output Cz. Over N steps,

$$z_{j+1} = Jz_j + Hv_j, \qquad \ell_t = Cz_t, \qquad \sum_{j=0}^{N-1} |v_j| \le V.$$

No Schur-stability assumption is required for this finite-horizon calculation. For $0 \le t \le N$ define $b_t = CJ^t z_0$, and for $1 \le t \le N$ let

$$g_t = \max_{0 \le k < t} |CJ^k H|.$$

An additional reserve $L_{\text{other}} \ge 0$ covers explicitly declared omitted risks. Bound cumulative execution costs by $c_{\text{exec}} \sum_{j<t} |v_j|$, with $c_{\text{exec}} \ge 0$. Both g_t and c_{exec} have monetary/quantity units; b_t, L_{other} and budget B have monetary units.

Proposition 1 (Finite-window quantity bound)**.** *For the absolute-output criterion, the worst attainable output-plus-cost envelope at each $t \ge 1$ over the stated quantity ball and cost class is*

$$|b_t| + L_{\text{other}} + (g_t + c_{\text{exec}})V.$$

For a one-sided signed loss criterion, replace $|b_t|$ by b_t. Let d_t denote the chosen background criterion. If some $d_t + L_{\text{other}} > B$, no nonnegative quantity cap certifies the full window. Otherwise, if at least one $g_t + c_{\text{exec}} > 0$, the largest cap under this contract is

$$V^* = \min_{\substack{1 \le t \le N \\ g_t + c_{\text{exec}} > 0}} \frac{B - L_{\text{other}} - d_t}{g_t + c_{\text{exec}}}.$$

Zero denominators impose no quantity restriction, provided their background already satisfies the budget. If all denominators vanish, this output contract places no quantity limit; other constraints may still apply. For a window lasting $\tau > 0$ seconds, V^*/τ is an average quantity-per-second cap, not a count of order messages.

Proof. Superposition gives

$$\ell_t = b_t + \sum_{j=0}^{t-1} CJ^{t-1-j} H v_j.$$

The triangle bound is $|b_t| + g_t V$. Its worst direction is attained by spending the whole quantity allowance in one pulse at a maximizing lag and choosing the sign to align with b_t; for one-sided loss choose the sign of the maximizing coefficient. A cost equal to its permitted envelope adds $c_{\text{exec}} V$. Thus these are exact suprema for the specified linear output/quantity/cost class. Imposing every timewise inequality, including the initial background, yields the minimum cap. A volume exceeding that minimum violates at least one constraint under an admissible pulse. □

Sharpness concerns this declared linear budget criterion. It does not authenticate the financial output, unknown impact map, fills, reserve, model error or noise coverage. The absolute-output criterion can be more conservative than actual one-sided dollar loss.

Proposition 2 (State coordinates and quantity units)**.** *For any invertible coordinate change T, transforming J, H, C, z_0 to $TJT^{-1}, TH, CT^{-1}, Tz_0$ leaves every b_t, g_t, V^* unchanged. Changing quantity units to $v'_j = av_j$, $a > 0$, requires $H' = H/a$ and $c'_{\text{exec}} = c_{\text{exec}}/a$; the numerical cap becomes aV^*, describing the same physical quantity.*

Proof. Cancel adjacent $T^{-1}T$ factors in $CT^{-1}(TJT^{-1})^t Tz_0$ and $CT^{-1}(TJT^{-1})^k TH$. Quantity conversion divides the monetary-per-quantity denominator by a. □

Proposition 3 (An identifiable financial functional despite cohort ambiguity)**.** *Suppose two reset linear models have the same price input/output transfer for the same calibrated executed-quantity input, possibly with different state dimensions. Suppose monetary loss in this experiment is a known fixed multiple of that price output, and both models use the same budget, cost and reserve. Then their finite-window quantity gates coincide, even if cohort identities or cohort count are not identified.*

Proof. Equality of the proper rational price transfers gives equality of every Laurent/Markov coefficient. Multiplying by the known monetary exposure preserves equality of the financial coefficients CJ^kH, so every g_t agrees. Reset states give $b_t = 0$ in both models. The cap formula therefore agrees. This uses no invertible correspondence between state spaces of different dimensions. □

An exact example compares one cohort of weight $1/5$ with two equal-rate cohorts of weights $1/10$ each, using $a = 1/2$, $\eta = 3/10$, $\kappa = 1$. Their price input responses coincide despite different state dimensions. The monetary output $C = (2, 0)$ or $(2, 0, 0)$ and input vector $(1, \eta)$ or $(1, \eta, \eta)$ give identical finite gates. Thus cohort ambiguity alone need not limit this financial functional. Unreset hidden states, unknown exposure/input maps, changing positions, fundamental risk and nonlinear marks can prevent this conclusion and require their own information.

Proposition 4 (Positive-cohort price gain depends on the input protocol)**.** *In the one-asset model with $0 \le a < 1$, $\kappa > 0$, $q_i \ge 0$, $0 < \eta_i < 1$ and $S = a + \kappa \sum_i q_i < 1$, let all cohort states filter the same price state, with update $y_{i,t+1} = (1 - \eta_i) y_{i,t} + \eta_i x_{t+1}$. A scalar executed input enters through $H = \kappa(1, \eta_1, \ldots, \eta_K)^\top$. For the known monetary price-only output $C = \lambda(1, 0, \ldots, 0)$,*

$$\sup_{k \ge 0} |CJ^kH| = |\lambda|\kappa, \qquad \sum_{k \ge 0} |CJ^kH| = \frac{|\lambda|\kappa}{1 - S}.$$

Thus the reset total-quantity pulse criterion and a sustained per-step input criterion use different gains. The first has no additional price-output impulse amplification beyond direct impact under these conditions; the second can grow large as S approaches one. This statement does not apply

without the positive, one-asset, fixed-output and stable-model assumptions, and does not include accumulated fees or fundamental/noise risks.

Proof. The price row of J sums to S, and cohort row i sums to $1 - \eta_i + \eta_i S < 1$. All entries are nonnegative, so $\|J\|_\infty < 1$ and $\|H\|_\infty = \kappa$. Every price-response coefficient is bounded by κ, with equality at lag zero. This gives the peak statement, including $|\lambda|$.

All price coefficients have the same sign after multiplying by λ. The sum is therefore the absolute DC gain. At the stationary response to a constant unit input, each cohort has $y_i = x$, and the price equation gives $x = Sx + \kappa$, hence $x = \kappa/(1 - S)$. Equivalently the sum is $|C(I - J)^{-1}H|$. □

The distinction also resolves a risk-ranking ambiguity. In the preceding economic supplement's positive-feedback example, increasing κ from 0.1 to 1 lowers all-state Euclidean initial-shock magnification from a value above 1.21 to 1, while the price-based executed-input peak increases from 0.1 to 1. With budget one and zero background/cost/reserve, its quantity cap falls from 10 to 1. Matrix-state amplification and physical execution risk answer different experiments; an uncalibrated state norm cannot substitute for the financial input/output gain. These are established positive-system/control deductions applied to this model, not new general control machinery or market validation.

2 Why a Window Cannot Be Reset without Carrying Its State

In the scalar model $z_{t+1} = 0.9z_t + v_t$, $C = H = 1$, start at zero with budget one and a one-step window. The first allowance is one. Resetting the second window to zero falsely permits another pulse of one, producing actual loss output 1.9. Carrying the actual state $z_1 = 1$ gives the second cap $1 - 0.9 = 0.1$ under the stated absolute-output contract. Window refreshes must use carried state, updated remaining capital, costs already paid and refreshed risk/model bounds. Repeating a single-window calculation is not an infinite-horizon solvency proof.

Displayed bid quantity does not supply a complete state embedding. For $H = (1, 10)^\top$, the price component of a unit pulse is one, while its Euclidean state norm is $\sqrt{101}$; a monetary output $C = (0, 1)$ responds by ten. The asserted bound of a full state norm by its price coordinate is false. A displayed top-five quantity and a price-impact coefficient also require a separate unit-consistent clearing relation before they determine H.

3 Regulatory Context and the Remaining Operational Gate

SEC Rule 15c3-5 already requires pre-trade credit/capital and erroneous-order controls; it is inaccurate to frame market-access protection as exclusively or inherently post-trade. See the SEC adopting release and staff guidance: adopting release, staff FAQ. The FAQ is staff guidance rather than the rule itself. The proposed mathematical quantity bound may inform a risk-control design under its assumptions; it does not establish regulatory compliance, welfare superiority or execution feasibility.

Operational use requires authenticated positions, marks, currencies, executions, fees, funding, margins and cancellations; justified input, state, model-error and noise bounds; and carried exposure/capital accounting. Empirical model rejections and reserve calibration remain unresolved.